You said:

Assess all the documentation and provide me a mental map of the required tasks
Do we need contextual or additional information?

Worked for 5m 44s

ChatGPT said:

You said:

what about the categorise function. why is the approach not a classifier? at some point in the document it asks what happens if the classifier degrades? it seems like a classifier ML problem

Worked for 2m 47s

ChatGPT said:

You said:

why do you consider a TFT-ID rather than a BM25 approach for instance? which design will have the highest likelihood to minimise degradation while scaling with more data ?
what if we use a local LLM to make the classification such as Bert or similar. aand we encapsulate the solution ina docker container. what are the potential gains ?

Worked for 3m 37s

ChatGPT said:

You said:

1 selection

Since we have an imbalanced dataset proper metrics shall be used for the classifier such us PR Curves. Shall we ask business what is the cost of an incorrect classification to prioritise the metric for the classifier or it is not necessary to ask business/stakeholders?

ChatGPT said:

You said:

Ok, do we have all the considerations to generate a lean design? or perhaps we are missing something?

Worked for 2m 9s

ChatGPT said:

You said:

MiniLM or Bert as part of the classification process which one is more suitable? what are the best prospects?

Worked for 56s

ChatGPT said:

You said:



image.png
Image

how to downloadd the mini?

Worked for 27s

ChatGPT said:

You said:

ok generate the project in ` zip file

Worked for 9m 20s

ChatGPT said:

You said:

Lests discuss the ingestion process from a DE Perspective, list the inferred schema of each file. If you can see the json file schema contains more information. My assumptions goes as

follows
1) In theory each file has solely info of an specific month
2) The merchant labels, what % has a representative similarity with the descriptions in the transaction files. so the model will derive, perhaps part of the MLOps involves HIL but we must minimise the probability, I beleive we create at first deterministic rules if its a 100% match, then similarity , MinLM and if the `CI is low then HIL. Is it reasonable to create a design based on a funnel classification?
3) How can I design my pipeline to be resilient against the new file? I have to make a big data approach in python despite I currentky handle small number of records.
4) Keep in mind that this pipeline must be production ready so what sort of guardrails I can create for a lean and robust design, consider data drift and data mutation, schema mutation might arise in the future

Worked for 9m 46s

ChatGPT said:

You said:

1 selection

Ok groceries is associated to TESCO MObile word , so what is the most likely category ? Mobile I guess

ChatGPT said:

You said:

We conclude that this file with the categories has some intuitive flaws, Ok, we can edit them and modify them. list all the potential flaws, potentia answer based on reason. Also, the 10 categories must have one more category, such us unknown , the category that demands HIL. Moreover do you think the funnel strategy with strict rules, else ML inference, else LLM inference else HIL works? As you know the number of categories will increase in the raw data over time. we want a pragmatic and realistic solution for a bank use case involving big data

Worked for 5m 10s

ChatGPT said:

You said:

1 selection

| `merchant_labels.csv` | Bill category for some merchant descriptors. Labelled by a person, so it may contain mistakes. | At some point in the MD file also ask what you dont trust in the data. I believe extending an Unknown category adds realism ans scalation. if not , how would you handle unassigned categories?

Worked for 1m 10s

ChatGPT said:

You said:

Ok I agree, we can add unknown and fundament the reason. Lets talk about the overall classification map. show me the overall classification approach. also we have to create a mlops or certain calibration or retraining without overengineering. what is your approach?

Worked for 1m 44s

ChatGPT said:

You said:

1 selection

in this architecture I do not see a conventional ML model, is that correct?

ChatGPT said:

You said:

1 selection

I think we have to decide the explicit models we will use in our funnel. if we propose a ML model, its asupervised model, do we have enough target labels in our potential training set? I believe we dont but you can object.

Worked for 2m 21s

ChatGPT said:

You said:

1 selection

are those steps achievable with the current data at hand? keep the design realistic

ChatGPT said:

You said:

Ok lets double check ig this Release 1 is aligned with all the points in the MD file.

Worked for 2m 23s

ChatGPT said:

You said:

1 selection

I disagree here. The MD explains that the company handles between 10 -20M customer per bank, this csv files are synthetic samples. Thus the overall architecture shall include a big data approach.

Worked for 2m 2s

ChatGPT said:

You said:

Ok, take this design as our baseline, lets discuss task 1. Previous EDA has defined obvious updates based on common logic. you proposed to keep the orignal raw file for lienage but make the proper correction in later stages. walk me through the arrpach for task 1

Worked for 3m 29s

ChatGPT said:

You said:

1 selection

EDA shall pay special attention to the date file, it has to be normalised. in some files the format is different.
At some point in the design you will generate a unique key for the cdc approach? please specify

Worked for 2m 25s

ChatGPT said:

You said:

Ok, do we have enugh clarity of shall we send some questions to ApTap? if so provide only relevant questions

Worked for 33s

ChatGPT said:

You said:

Ok are we ready to generate the project file based on the incremental approach? are we ready to cover most of the deployment?

Worked for 52s

ChatGPT said:

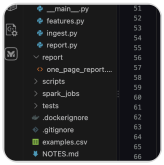
You said:

Ok lets proceed with the new release 1

Worked for 20m 56s

ChatGPT said:

You said:



Screenshot 2026...

Image

1 selection

I do not see the path pipeline data in the folder project files. lets create a sequential set of script to run in the project all the critical proceses .venv ...

Worked for 9m 18s

ChatGPT said:

You said:

1 selection

where are these tables located in the project or are they dynamically generated, even so, how to identify them and track them

Worked for 4m 35s

ChatGPT said:

You said:



Screenshot 2026...
Image

I understand I can use the SQLite library and query the tables. still shall I use an add on, which one?

ChatGPT said:

You said:

lets focus on the current local solution, all the bash codes cover the local solution for the pipeline ingestion , correct?

ChatGPT said:

You said:

explain me the script execution here. the second parameter engine, what is it ,
#!/usr/bin/env bash
set -euo pipefail
source "$(dirname "${BASH_SOURCE[0]}")/_common.sh"
need_venv
as_of="${1:-2026-09-30}"
engine="${2:-lexical}"
heading "03 | Incremental ingest -> canonical Silver -> classification -> bills/switches -> report"
"$VENV_PY" -m pipeline data/ --engine "$engine" --as-of "$as_of"

ChatGPT said:

You said:

(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 % bash run_all.sh

===== ApTap Release 1 | complete local workflow =====

===== 01 | Read-only EDA: date formats, source periods and SHA-256 =====
transactions_2026_07.csv [csv_v1]: 43,415 records, booking months {'2026-07': 43415}, issues {}
transactions_2026_08.csv [csv_v1]: 46,790 records, booking months {'2026-07': 1052, '2026-08': 45738}, issues {'cross_period_booking_date': 1052}
transactions_2026_09.jsonl [jsonl_v2]: 44,950 records, booking months {'2026-09': 44950}, issues {}
Input EDA: outputs/source_eda.json

===== 02 | Unit and incremental/CDC regression tests =====
.....
[100%]
12 passed in 0.10s

===== 03 | Incremental ingest -> canonical Silver -> classification -> bills/switches -> report =====
RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'COMMITTED'),

('transactions_2026_08.csv', 'COMMITTED'),
('transactions_2026_09.jsonl', 'COMMITTED')]
Unknown: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Runtime 6.719s; peak RSS 57.2 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html

===== 04 | Standalone merchant categoriser (input + category output + reviews) =====
Wrote outputs/examples_categorised.csv; unresolved occurrences 2 (internal category unknown)

===== 05 | Exploratory leave-one-description-out classifier benchmark =====
char_tfidf 0.789 0.635

Benchmark: outputs/classifier_benchmark.json

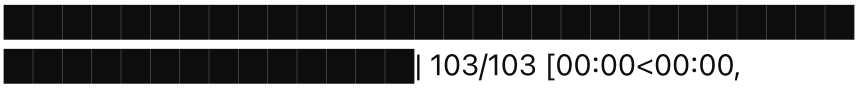
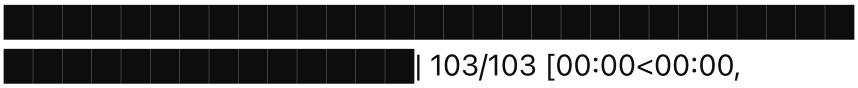
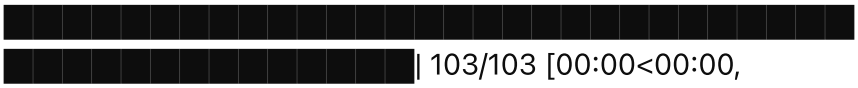
===== 06 | Verify rerun idempotency, checkpoint and 4 required CSV hashes =====
RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_08.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_09.jsonl', 'SKIPPED_IDENTICAL')]
Unknown: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Runtime 1.638s; peak RSS 60.4 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html
Rerun verification: outputs/workflow_rerun_verification.json
PASS: 4 CSVs byte-identical; 0 new batches

SUCCESS. Required CSVs are in outputs/; one-page report in report/.
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %
I guess the process was executed immediately since the SQLite tables already had the incrementals right?
The second step is download the MiniLM correct? if not provide guidance in further steps

ChatGPT said:

You said:

1 selection

```
===== 09 | OPTIONAL offline MiniLM inference and benchmark =====
Loading weights: 100%|

| 103/103 [00:00<00:00, 5362.36it/s]
RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_08.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_09.jsonl', 'SKIPPED_IDENTICAL')]
Unknown: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Runtime 7.518s; peak RSS 524.7 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html
Loading weights: 100%|

| 103/103 [00:00<00:00, 9629.61it/s]
Wrote outputs/examples_minilm.csv; unresolved occurrences 2 (internal category unknown)
Loading weights: 100%|

| 103/103 [00:00<00:00, 8150.58it/s]
char_tfidf 0.789 0.635
minilm 0.754 0.657
```

Worked for 1m 39s

ChatGPT said:

You said:

1 selection

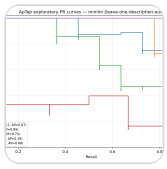
I think that something must be added, this classification task is based on imbalanced data, shouldnt be consider PR curves as a meaninful or perhaps key ratio? how are you obtaining the F1 fir

instance. I understand we are not using supervised models since we do not have a testing set

Worked for 5m 2s

ChatGPT said:

You said:



pr_curves_minil...
Image

pr_evaluation_re...
Code

pr_curves_char_t...
Image

Pr By Category

.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 % python scripts/benchmark_pr_curves.py
char_tfidf: top1_accuracy=0.789, macro_F1=0.635, macro_AP=0.698
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs/pr_evaluation
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 % HF_HUB_OFFLINE=1 python scripts/benchmark_pr_curves.py --with-minilm
Loading weights: 100%|
103/103 [00:00<00:00, 4930.81it/s]

char_tfidf: top1_accuracy=0.789, macro_F1=0.635, macro_AP=0.698
minilm: top1_accuracy=0.754, macro_F1=0.657, macro_AP=0.654
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs/pr_evaluation
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %

Worked for 1m 32s

Analyzed

```
import os,json,pandas as pd
paths=['/mnt/data/pr_by_category.csv','/mnt/data/pr_evaluation_report.json']
for p in paths: print(p,os.path.exists(p),os.path.getsize(p))
df=pd.read_csv(paths[0]);print(df.columns.tolist());print(df.to_string(index=False))
d=json.load(open(paths[1]));
print(d.keys());print([(m.get("model"),m.get("top1_accuracy"),m.get("top1_macro_f1"),m.get("macro_average_precision")) for m in d['models']])
```

STDOUT/STDERR

```
/mnt/data/pr_by_category.csv True 1477
/mnt/data/pr_evaluation_report.json True 7770
['model', 'category', 'support', 'prevalence', 'average_precision', 'top1_pr
     model       category  support  prevalence  average_precision  top1_prec
char_tfidf      broadband       11     0.19298            0.88497         0.
char_tfidf  car_insurance        3     0.05263            0.63889         0.
char_tfidf    council_tax        3     0.05263            0.86667         1.
char_tfidf         energy        9     0.15789            0.90889         1.
char_tfidf            gym        1     0.01754            0.01754         0.
char_tfidf home_insurance        2     0.03509            0.32500         0.
char_tfidf         mobile       11     0.19298            0.91338         0.
char_tfidf     not_a_bill        6     0.10526            0.48621         0.
char_tfidf      streaming        7     0.12281            0.93793         0.
char_tfidf          water        4     0.07018            1.00000         0.
    minilm      broadband       11     0.19298            0.86827         0.
    minilm  car_insurance        3     0.05263            0.76667         0.
    minilm    council_tax        3     0.05263            1.00000         1.
    minilm         energy        9     0.15789            0.89192         1.
    minilm            gym        1     0.01754            0.01754         0.
    minilm home_insurance        2     0.03509            0.08333         0.
    minilm         mobile       11     0.19298            0.74616         0.
    minilm     not_a_bill        6     0.10526            0.28384         0.
    minilm      streaming        7     0.12281            0.87987         1.
```

```
    minilm          water        4     0.07018            1.00000         1.
dict_keys(['design', 'caveats', 'models'])
[('char_tfidf', 0.78947, 0.6355, 0.69795), ('minilm', 0.75439, 0.65702, 0.65
```

ChatGPT said:

‹/›

You said:

Ok what is the next step to finish this implementation?

Worked for 2m 3s

ChatGPT said:

You said:

1 selection

Descriptions requiring review: 21
CARD PAYMENT TO JOHN S Candidate: water
Method: lexical_minilm_disagree
LIVERPOOL VICTORIA Candidate: not_a_bill
Method: lexical_minilm_disagree
DIRECT LINE INS Candidate: home_insurance
Method: lexical_minilm_agree
VIRGIN MEDIA Candidate: broadband Method:
lexical_minilm_agree
LV INSURANCE Candidate: home_insurance
Method: lexical_minilm_agree
AVIVA DD Candidate: car_insurance Method:
lexical_minilm_agree
AMZNPRIME UK Candidate: streaming Method:
lexical_minilm_agree
AMAZON PRIME* Candidate: streaming Method:
lexical_minilm_agree
DIRECTLINE DD Candidate: home_insurance
Method: lexical_minilm_agree
AVIVA INSURANCE Candidate: car_insurance
Method: lexical_minilm_agree
BRITISH GAS SERVICES Candidate: energy
Method: lexical_minilm_agree
CROYDON COUNCIL Candidate: council_tax
Method: lexical_minilm_agree

LV= DD Candidate: home_insurance Method: lexical_minilm_agree
SKY DIGITAL Candidate: broadband Method: lexical_minilm_agree
VIRGIN MEDIA PYMTS Candidate: broadband Method: lexical_minilm_agree
SKY UK LIMITED DD Candidate: broadband Method: lexical_minilm_agree
VIRGINMEDIA DD Candidate: broadband Method: lexical_minilm_agree
BT GROUP PLC Candidate: gym Method: lexical_minilm_disagree
SKY UK LTD Candidate: broadband Method: lexical_minilm_agree
EE LIMITED Candidate: mobile Method: lexical_minilm_agree
VODAFONE LTD Candidate: mobile Method: lexical_minilm_agree
(.venv) benja

one step at a time for analysisi

ChatGPT said:

You said:

1 selection

I did not create it, I believe I can use a python script to do so without the extension

ChatGPT said:

You said:

Merchant: CARD PAYMENT TO JOHN S
---
transactions : 8605
customers : 1493
distinct_amounts : 6901
first_date : 2026-07-01
last_date : 2026-09-30
debit_transactions : 8605

Monthly distribution:
2026-07: 2836
2026-08: 2888
2026-09: 2881

ChatGPT said:

You said:

On , but what are you trying to do, do you want me to make an in depth assessment of each of the flagged transactions?

ChatGPT said:

You said:

Ok, proceed

Worked for 7m 7s

ChatGPT said:

ApTap_merchant_audit_all_92.csv
Spreadsheet

ApTap_merchant_review_proposals.csv
Spreadsheet

ApTap_MERCHANT_GOVERNANCE_AUDIT.md
Document

You said:

1 cell

1 selection

have you updated the requirements.txt I believeyou need matplotlib and other package

Worked for 4m 7s

ChatGPT said:

You said:

why do we have so many requirements files, is this an effective and lean project structure in vscode?

Worked for 1m 3s

ChatGPT said:

You said:

provide me the improved version

Worked for 4m 18s

ChatGPT said:

You said:

```
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %
python -m pip install -r requirements.txt
  bash scripts/workflows/02_run_tests.sh
  bash scripts/workflows/14_evaluate_pr_curves.sh

Requirement already satisfied: pytest<10,>=8 in
./.venv/lib/python3.14/site-packages (from -r requirements.txt
(line 4)) (9.1.1)
Requirement already satisfied: numpy<3,>=1.26 in
./.venv/lib/python3.14/site-packages (from -r requirements.txt
(line 5)) (2.5.4)
Requirement already satisfied: scikit-learn<2,>=1.5 in
./.venv/lib/python3.14/site-packages (from -r requirements.txt
(line 6)) (1.9.1)
Requirement already satisfied: matplotlib<4,>=3.8 in
./.venv/lib/python3.14/site-packages (from -r requirements.txt
(line 7)) (3.11.2)
Requirement already satisfied: iniconfig>=1.0.1 in
./.venv/lib/python3.14/site-packages (from pytest<10,>=8->-r
requirements.txt (line 4)) (2.3.1)
Requirement already satisfied: packaging>=22 in
./.venv/lib/python3.14/site-packages (from pytest<10,>=8->-r
requirements.txt (line 4)) (26.3)
Requirement already satisfied: pluggy<2,>=1.5 in
./.venv/lib/python3.14/site-packages (from pytest<10,>=8->-r
requirements.txt (line 4)) (1.6.0)
Requirement already satisfied: pygments>=2.7.2 in
```

./.venv/lib/python3.14/site-packages (from pytest<10,>=8->-r requirements.txt (line 4)) (2.21.0)
Requirement already satisfied: scipy>=1.10.0 in ./.venv/lib/python3.14/site-packages (from scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (1.18.1)
Requirement already satisfied: joblib>=1.4.0 in ./.venv/lib/python3.14/site-packages (from scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (1.6.0)
Requirement already satisfied: narwhals>=2.0.1 in ./.venv/lib/python3.14/site-packages (from scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (2.27.1)
Requirement already satisfied: threadpoolctl>=3.5.0 in ./.venv/lib/python3.14/site-packages (from scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (3.7.0)
Requirement already satisfied: contourpy>=1.0.1 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (1.4.0)
Requirement already satisfied: cycler>=0.10 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (4.67.0)
Requirement already satisfied: kiwisolver>=1.3.1 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (1.5.1)
Requirement already satisfied: pillow>=9 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (12.3.0)
Requirement already satisfied: pyparsing>=3 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (2.9.0.post0)
Requirement already satisfied: cloudpickle>=3.0 in ./.venv/lib/python3.14/site-packages (from joblib>=1.4.0->scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (3.1.2)
Requirement already satisfied: six>=1.5 in ./.venv/lib/python3.14/site-packages (from python-dateutil>=2.7->matplotlib<4,>=3.8->-r requirements.txt (line 7)) (1.17.0)

[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: pip install --upgrade pip

```
===== 02 | Full local regression suite (ingestion, CDC,
governance, packaging) =====
F.F.....                                                  [100%]
=============================================
FAILURES
=============================================
_____________________________________
test_pr_evaluation_is_packaged
_____________________________________

   def test_pr_evaluation_is_packaged():
>      assert (ROOT / "scripts/benchmark_pr_curves.py").is_file()
E      AssertionError: assert False
E       +  where False = is_file()
E       +    where is_file =
(PosixPath('/Users/benjaminps/BWalker/Projects/UK JS/aptap-
v1') / 'scripts/benchmark_pr_curves.py').is_file

tests/test_evaluation_packaging.py:8: AssertionError
_____________________________
test_setup_and_optional_scripts_use_correct_profiles
_____________________________

   def test_setup_and_optional_scripts_use_correct_profiles():
       setup = (ROOT /
"scripts/workflows/00_setup_venv.sh").read_text()
       optional = (ROOT /
"scripts/workflows/08_optional_install_minilm.sh").read_text()
>      assert "-r requirements.txt" in setup
E      AssertionError: assert '-r requirements.txt' in '#!/usr/bin/env
bash\n# Usage: bash scripts/workflows/00_setup_venv.sh [--
offline]\n# --offline skips the optional pyt...hich always use
.venv/bin/python directly.\nNo MiniLM weights and no Spark
packages are installed by this step.\nMSG\n'

tests/test_evaluation_packaging.py:36: AssertionError
==================================== short test
summary info
=====================================
FAILED
tests/test_evaluation_packaging.py::test_pr_evaluation_is_packa
ged - AssertionError: assert False
FAILED
tests/test_evaluation_packaging.py::test_setup_and_optional_sc
ripts_use_correct_profiles - AssertionError: assert '-r
```

requirements.txt' in '#!/usr/bin/env bash\n# Usage: bash scripts/workf...
2 failed, 17 passed in 0.16s
bash: scripts/workflows/14_evaluate_pr_curves.sh: No such file or directory
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %

Worked for 3m 2s

ChatGPT said:

You said:

with the current design , what is the rate of effcetiveness of the classifier, what percentage require manual review?

ChatGPT said:

You said:

feedback answers from the employer
questions:
1. Transaction identity and CDC: Are `transaction_id` values guaranteed to remain unique and stable within a bank, including after platform migrations? Can an existing transaction be resent with corrected values, and if so, how should we identify the authoritative version?
2. September JSONL semantics: Can we assume `value` is always a non-negative magnitude, `direction` determines its sign, and GBP is the implicit currency unless explicitly provided otherwise?
3. Unknown classification: The brief specifies ten categories. Would you accept an additional `unknown` outcome for descriptors that cannot be reliably classified, allowing us to route uncertain cases to human review instead of forcing a potentially incorrect classification?
4. October incremental processing: When the October file is added, should `recurring_bills.csv` continue representing bills active at the end of September, or advance to October? Our design can support either through a configurable reporting date.
Answers
1. A transaction_id identifies one transaction at this bank, and the bank kept the same IDs through a migration. However, the same

ID can appear more than once in the files. Yes, the bank can resend a transaction it has already sent, and sometimes the resent copy has corrected values. The files have no version number or "last updated" field. How you decide which version to keep is your call. Tell us what you chose and why, and make sure the rows you don't keep are logged.

2. September values. Yes. value is always a positive amount, direction gives the sign, and every value is in GBP. The formatting of value can vary, so don't rely on one fixed format.

3. An "unknown" category. The category column has to contain one of the ten categories, because we score it automatically against those. We like the idea of sending uncertain cases to a person, though. You're welcome to add an extra column, such as a confidence score or a needs_review flag, and to explain in your notes how that review would work in production.

4. Reporting date after October. Move it forward. When the October file is included, recurring_bills.csv should show the bills being paid at the end of October (31 October 2026). A configurable date is fine; just tell us how to set it.\

Worked for 2m 43s

ChatGPT said:

You said:

can you perform all the updates at once or will you prefer an incremental approach

ChatGPT said:

You said:

1 selection

I prefer an optimised band new release and I will run everything from the start

Worked for 11m 16s

ChatGPT said:

You said:

.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 % #
Step 1: Create Python virtual environment
bash scripts/workflows/00_setup_venv.sh

Step 2: Activate the environment
source .venv/bin/activate

Step 3: Execute the complete workflow
bash run_all.sh
zsh: command not found: #

===== 00 | Create or reuse .venv =====
Python 3.14.3

===== Install local test + PR evaluation dependencies =====
Requirement already satisfied: pytest<10,>=8 in ./.venv/lib/python3.14/site-packages (from -r requirements.txt (line 4)) (9.1.1)
Requirement already satisfied: numpy<3,>=1.26 in ./.venv/lib/python3.14/site-packages (from -r requirements.txt (line 5)) (2.5.4)
Requirement already satisfied: scikit-learn<2,>=1.5 in ./.venv/lib/python3.14/site-packages (from -r requirements.txt (line 6)) (1.9.1)
Requirement already satisfied: matplotlib<4,>=3.8 in ./.venv/lib/python3.14/site-packages (from -r requirements.txt (line 7)) (3.11.2)
Requirement already satisfied: iniconfig>=1.0.1 in ./.venv/lib/python3.14/site-packages (from pytest<10,>=8->-r requirements.txt (line 4)) (2.3.1)
Requirement already satisfied: packaging>=22 in ./.venv/lib/python3.14/site-packages (from pytest<10,>=8->-r requirements.txt (line 4)) (26.3)
Requirement already satisfied: pluggy<2,>=1.5 in ./.venv/lib/python3.14/site-packages (from pytest<10,>=8->-r requirements.txt (line 4)) (1.6.0)
Requirement already satisfied: pygments>=2.7.2 in ./.venv/lib/python3.14/site-packages (from pytest<10,>=8->-r requirements.txt (line 4)) (2.21.0)
Requirement already satisfied: scipy>=1.10.0 in

./.venv/lib/python3.14/site-packages (from scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (1.18.1)
Requirement already satisfied: joblib>=1.4.0 in ./.venv/lib/python3.14/site-packages (from scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (1.6.0)
Requirement already satisfied: narwhals>=2.0.1 in ./.venv/lib/python3.14/site-packages (from scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (2.27.1)
Requirement already satisfied: threadpoolctl>=3.5.0 in ./.venv/lib/python3.14/site-packages (from scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (3.7.0)
Requirement already satisfied: contourpy>=1.0.1 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (1.4.0)
Requirement already satisfied: cycler>=0.10 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (4.67.0)
Requirement already satisfied: kiwisolver>=1.3.1 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (1.5.1)
Requirement already satisfied: pillow>=9 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (12.3.0)
Requirement already satisfied: pyparsing>=3 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in ./.venv/lib/python3.14/site-packages (from matplotlib<4,>=3.8->-r requirements.txt (line 7)) (2.9.0.post0)
Requirement already satisfied: cloudpickle>=3.0 in ./.venv/lib/python3.14/site-packages (from joblib>=1.4.0->scikit-learn<2,>=1.5->-r requirements.txt (line 6)) (3.1.2)
Requirement already satisfied: six>=1.5 in ./.venv/lib/python3.14/site-packages (from python-dateutil>=2.7->matplotlib<4,>=3.8->-r requirements.txt (line 7)) (1.17.0)

[notice] A new release of pip is available: 25.3 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
Environment ready. In this Terminal, optional activation:
  source .venv/bin/activate
Or run the numbered scripts, which always use .venv/bin/python directly.

No MiniLM weights and no Spark packages are installed by this step.
zsh: command not found: #
zsh: command not found: #

===== ApTap Release 2 | end-to-end from clean inputs =====

===== 01 | Read-only EDA: date formats, source periods and SHA-256 =====
transactions_2026_07.csv [csv_v1]: 43,415 records, booking months {'2026-07': 43415}, issues {}
transactions_2026_08.csv [csv_v1]: 46,790 records, booking months {'2026-07': 1052, '2026-08': 45738}, issues {'cross_period_booking_date': 1052}
transactions_2026_09.jsonl [jsonl_v2]: 44,950 records, booking months {'2026-09': 44950}, issues {}
Input EDA: outputs/source_eda.json

===== 03 | Incremental ingest -> canonical Silver -> classification -> bills/switches -> report =====
RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'COMMITTED'), ('transactions_2026_08.csv', 'COMMITTED'), ('transactions_2026_09.jsonl', 'COMMITTED')]
Flagged for review: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Runtime 7.678s; peak RSS 82.8 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html

===== 04 | Standalone merchant categoriser (input + category output + reviews) =====
Wrote outputs/examples_categorised.csv; unresolved occurrences 2 (provisional ten-class predictions, with review flags)

===== 05 | Exploratory leave-one-description-out classifier benchmark =====
char_tfidf 0.789 0.635

Benchmark: outputs/classifier_benchmark.json

===== 12 | Audit 92 merchant descriptors and produce read-only governance proposals =====

AUDITED 92 descriptors, 133,692 clean transactions
REVIEW REQUIRED 21 descriptions / 15,033 transaction occurrences
ADDITIONAL RISK 2 accepted demo mappings / 851 transactions
CONSERVATIVE PROPOSAL 23 review-required / 15,884 transactions
NO NEW AUTO-APPROVALS. Proposals written to outputs/merchant_governance

===== 13 | Compare current demo registry with conservative proposal without changing SQLite =====
BASELINE {'bill_candidates': 6681, 'switch_candidates': 209, 'review_descriptors': 21}
CONSERVATIVE {'bill_candidates': 6401, 'switch_candidates': 209, 'review_descriptors': 23}
DELTA {'bill_candidates': -280, 'switch_candidates': 0}
No source data or curated aliases modified.

===== 14 | Exploratory PR curves, per-category AP and top-1 F1 =====
char_tfidf: top1_accuracy=0.789, macro_F1=0.635, macro_AP=0.698
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs/pr_evaluation
Results: outputs/pr_evaluation/

===== 02 | Full local regression suite (ingestion, CDC, governance, packaging) =====
..... [100%]
24 passed in 0.50s

===== 06 | Verify rerun idempotency, checkpoint and 4 required CSV hashes =====
RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_08.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_09.jsonl', 'SKIPPED_IDENTICAL')]
Flagged for review: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Runtime 1.75s; peak RSS 60.9 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html
Rerun verification: outputs/workflow_rerun_verification.json

PASS: 4 CSVs byte-identical; 0 new batches

SUCCESS. Required CSVs are in outputs/; one-page report in report/.
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %
Since I started from scratch I have to create the MinLM model too

ChatGPT said:

You said:

Pasted text.txt
Document

ChatGPT said:

You said:

what is the next step?

Worked for 47s

ChatGPT said:

You said:

ok what nexxt step?

Worked for 49s

ChatGPT said:

You said:

proceed

Worked for 9m 21s

ChatGPT said:

You said:

Ok what do do, what is the proposed action

Worked for 1m 18s

ChatGPT said:

You said:

1 selection

I executed this script but I have to create the model again and run the classifier using the new model I believe so

Worked for 25s

ChatGPT said:

You said:

do you see the list of conditional rues in your workflow. we have 2 ifs (approved merchant alias) and trusted evidence. is this logic executed in which script. isnt is suppored to be activated every time I want to classify? does step 2 complies with all that conditional logic?

Worked for 1m 24s

ChatGPT said:

You said:

where do I see the classification? in which outputs document

ChatGPT said:

You said:



image(1).png
Image

lets assess this file there is a column of risk, this one is generated through the average of which predictions? I need to identify where in the program is that logic and how it behaves

Worked for 1m 22s

ChatGPT said:

You said:

Ok, so far I have no further questions what else do we need to comply with the md file in the Tap assignment

Worked for 2m 23s

ChatGPT said:

You said:

1 selection

what is the scope of commercial report? can we provide more meaningful information?

Worked for 3m 14s

ChatGPT said:

You said:

what would you think about an additional funnel report I imagine taking a month as a filter raw transactions vs effective transactions and the % of classification filtered by risk. In theory the higher is the one that needs review. I believe is reasonable to ask what is the impact of those high risk classifications

Worked for 2m 51s

ChatGPT said:

You said:

why are we using html? because it is the simplest approach I guess. if we use something else? angular or something more colourful? and interactive? do not generate updates, just discuss it

ChatGPT said:

You said:

1 selection

since we have aligned the insights lets use this option. do we need to generate everything from ground zero or des it involve a medium risk patch

ChatGPT said:

You said:

Ok, before you do this, Lets go to the ML/AI inference stage in this project. we have proposed a inference based on TDF MinlLM models right? ar wich point we compare both results obtained for both preditctions? In which percentages the inference is based in the MinLM? do you think if we upgrade to a more robutst LLM we can minimize the high risk records to review. minimise HIL and maximise automation or perhaps with the current info is unlikely to obtain a higher automation in the classification process

Worked for 2m 31s

ChatGPT said:

You said:

1 selection

can you align this new approach with the updated report? what is the impact?

Worked for 2m 5s

ChatGPT said:

You said:

Ok, what else shall we do to finish this assessment?

Worked for 1m 16s

ChatGPT said:

You said:

1 selection

Ok what do you need to execute this implementation? are you planning to provide me the latest update and start based on that?

Worked for 47s

ChatGPT said:

You said:

ok I have updated the files , provide bash code sequence to run everything. from scratch

Worked for 2m 8s

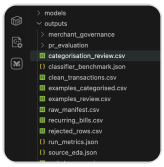
ChatGPT said:

You said:

I realise that the output file already contain information, if I delete all the info , the idea is that running all the processes will generate everything again right?

Stopped thinking

You said:



Screenshot 2026...
Image

I realise that the output file already contain information, if I delete all the info , the idea is that running all the processes will generate everything again right?

ChatGPT said:

You said:

1 selection

where is it? exact path

ChatGPT said:

You said:

Ok everything has been generated, what are the next steps one by one to complete the assignment

Worked for 1m 25s

ChatGPT said:

You said:

APTAP — RECURRING BILL VALIDATION

Reporting date: 2026-09-30
Total candidates: 6681
Unique bill keys: 6681
Duplicate bill keys: 0
Non-positive amounts: 0

Cadence distribution:
  monthly: 6561
  monthly_candidate: 120

Category distribution:
  mobile: 1246
  energy: 1173
  streaming: 988
  water: 881
  broadband: 856
  council_tax: 796
  gym: 461
  car_insurance: 280

Next expected payment month:
  2026-10: 6681

Worked for 2m 26s

ChatGPT said:

You said:

Total bills: 6681
Bills with multiple monthly payments: 267
Bills with amount difference >10%: 7

Largest differences:
('C00164', 'octopus', 'energy') | Reported: £127.73 | Monthly median: £191.60 | Difference: 33.3%
('C00658', 'eon', 'energy') | Reported: £114.67 | Monthly median: £172.00 | Difference: 33.3%
('C00918', 'octopus', 'energy') | Reported: £78.12 | Monthly median: £117.18 | Difference: 33.3%
('C01309', 'octopus', 'energy') | Reported: £150.02 | Monthly median: £225.03 | Difference: 33.3%
('C00707', 'ovo', 'energy') | Reported: £109.72 | Monthly median: £122.89 | Difference: 10.7%
('C00710', 'ovo', 'energy') | Reported: £163.55 | Monthly median: £183.18 | Difference: 10.7%
('C01241', 'edf', 'energy') | Reported: £80.42 | Monthly median: £90.07 | Difference: 10.7%
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %

Worked for 2m 16s

ChatGPT said:

You said:

Ok, proceed as recommended. what are the next steps

Worked for 7m 34s

ChatGPT said:

You said:

1 selection

(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %
bash run_all.sh

===== ApTap Release 2 | end-to-end from clean inputs =====

===== 01 | Read-only EDA: date formats, source periods and SHA-256 =====

transactions_2026_07.csv [csv_v1]: 43,415 records, booking months {'2026-07': 43415}, issues {}
transactions_2026_08.csv [csv_v1]: 46,790 records, booking months {'2026-07': 1052, '2026-08': 45738}, issues {'cross_period_booking_date': 1052}
transactions_2026_09.jsonl [jsonl_v2]: 44,950 records, booking months {'2026-09': 44950}, issues {}
Input EDA: outputs/source_eda.json

===== 03 | Incremental ingest -> canonical Silver -> classification -> bills/switches -> report =====
RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_08.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_09.jsonl', 'SKIPPED_IDENTICAL')]
Flagged for review: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Bill evidence verified: PASSED | {'RECENT_CONSECUTIVE_MONTHS': 6561, 'HISTORICAL_CONSECUTIVE_ONLY': 0, 'MONTHLY_CANDIDATE_INSUFFICIENT_HISTORY': 120} | monthly-total outliers: 7
Switch evidence verified: PASSED | {'PROVISIONAL': 87, 'REPEATED_NEW_PROVIDER': 122, 'SUPPORTED_PAYMENT_PATTERN': 0} | externally confirmed: 0
Runtime 3.498s; peak RSS 70.5 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html

===== 04 | Standalone merchant categoriser (input + category output + reviews) =====
Wrote outputs/examples_categorised.csv; unresolved occurrences 2 (provisional ten-class predictions, with review flags)

===== 05 | Exploratory leave-one-description-out classifier benchmark =====
char_tfidf 0.789 0.635

Benchmark: outputs/classifier_benchmark.json

===== 12 | Audit 92 merchant descriptors and produce read-only governance proposals =====

AUDITED 92 descriptors, 133,692 clean transactions
REVIEW REQUIRED 21 descriptions / 15,033 transaction occurrences
ADDITIONAL RISK 2 accepted demo mappings / 851 transactions
CONSERVATIVE PROPOSAL 23 review-required / 15,884 transactions
NO NEW AUTO-APPROVALS. Proposals written to outputs/merchant_governance

===== 13 | Compare current demo registry with conservative proposal without changing SQLite =====
BASELINE {'bill_candidates': 6681, 'switch_candidates': 209, 'review_descriptors': 21}
CONSERVATIVE {'bill_candidates': 6401, 'switch_candidates': 209, 'review_descriptors': 23}
DELTA {'bill_candidates': -280, 'switch_candidates': 0}
No source data or curated aliases modified.

===== 14 | Exploratory PR curves, per-category AP and top-1 F1 =====
char_tfidf: top1_accuracy=0.789, macro_F1=0.635, macro_AP=0.698
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs/pr_evaluation
Results: outputs/pr_evaluation/

===== 02 | Full local regression suite (ingestion, CDC, governance, packaging) =====
....
[100%]
50 passed in 0.56s

===== 06 | Verify rerun idempotency, checkpoint and 4 required CSV hashes =====
RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_08.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_09.jsonl', 'SKIPPED_IDENTICAL')]
Flagged for review: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Bill evidence verified: PASSED | {'RECENT_CONSECUTIVE_MONTHS': 6561, 'HISTORICAL_CONSECUTIVE_ONLY': 0,

'MONTHLY_CANDIDATE_INSUFFICIENT_HISTORY': 120} | monthly-total outliers: 7
Switch evidence verified: PASSED | {'PROVISIONAL': 87, 'REPEATED_NEW_PROVIDER': 122, 'SUPPORTED_PAYMENT_PATTERN': 0} | externally confirmed: 0
Runtime 3.097s; peak RSS 72.6 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html
Rerun verification: outputs/workflow_rerun_verification.json
PASS: 4 CSVs byte-identical; 0 new batches

SUCCESS. Required CSVs are in outputs/; one-page report in report/.
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %

ChatGPT said:

You said:

ok proceed and provide me the patch code

Worked for 19m 56s

ChatGPT said:

You said:

(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 % bash run_all.sh

===== ApTap Release 2 | end-to-end from clean inputs =====

===== 01 | Read-only EDA: date formats, source periods and SHA-256 =====
transactions_2026_07.csv [csv_v1]: 43,415 records, booking months {'2026-07': 43415}, issues {}
transactions_2026_08.csv [csv_v1]: 46,790 records, booking months {'2026-07': 1052, '2026-08': 45738}, issues {'cross_period_booking_date': 1052}
transactions_2026_09.jsonl [jsonl_v2]: 44,950 records, booking months {'2026-09': 44950}, issues {}
Input EDA: outputs/source_eda.json

===== 03 | Incremental ingest -> canonical Silver -> classification -> bills/switches -> report =====

RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_08.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_09.jsonl', 'SKIPPED_IDENTICAL')]
Flagged for review: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Bill evidence verified: PASSED | {'RECENT_CONSECUTIVE_MONTHS': 6561, 'HISTORICAL_CONSECUTIVE_ONLY': 0, 'MONTHLY_CANDIDATE_INSUFFICIENT_HISTORY': 120} | monthly-total outliers: 7
Switch evidence verified: PASSED | {'PROVISIONAL': 87, 'REPEATED_NEW_PROVIDER': 122, 'SUPPORTED_PAYMENT_PATTERN': 0} | externally confirmed: 0
Runtime 3.758s; peak RSS 72.4 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html

===== 04 | Standalone merchant categoriser (input + category output + reviews) =====
Wrote outputs/examples_categorised.csv; unresolved occurrences 2 (provisional ten-class predictions, with review flags)

===== 05 | Exploratory leave-one-description-out classifier benchmark =====
char_tfidf 0.789 0.635

Benchmark: outputs/classifier_benchmark.json

===== 12 | Audit 92 merchant descriptors and produce read-only governance proposals =====
AUDITED 92 descriptors, 133,692 clean transactions
REVIEW REQUIRED 21 descriptions / 15,033 transaction occurrences
ADDITIONAL RISK 2 accepted demo mappings / 851 transactions
CONSERVATIVE PROPOSAL 23 review-required / 15,884 transactions
NO NEW AUTO-APPROVALS. Proposals written to outputs/merchant_governance

===== 13 | Compare current demo registry with conservative proposal without changing SQLite =====

BASELINE {'bill_candidates': 6681, 'switch_candidates': 209, 'review_descriptors': 21}
CONSERVATIVE {'bill_candidates': 6401, 'switch_candidates': 209, 'review_descriptors': 23}
DELTA {'bill_candidates': -280, 'switch_candidates': 0}
No source data or curated aliases modified.

===== 14 | Exploratory PR curves, per-category AP and top-1 F1 =====
char_tfidf: top1_accuracy=0.789, macro_F1=0.635, macro_AP=0.698
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs/pr_evaluation
Results: outputs/pr_evaluation/

===== 15 | Offline commercial and inference-risk reports (read-only) =====
DASHBOARD: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/report/risk_dashboard.html | 3 months | 92descriptors | MiniLM diagnostics=False | zero external resources
One-page commercial report: report/one_page_report.html
Interactive dashboard: report/risk_dashboard.html

===== 02 | Full local regression suite (ingestion, CDC, governance, packaging) =====
.....
[100%]
54 passed in 0.59s

===== 06 | Verify rerun idempotency, checkpoint and 4 required CSV hashes =====
RECONCILIATION: 135,155 = 133,692 clean + 1,463 rejected
Batch actions: [('transactions_2026_07.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_08.csv', 'SKIPPED_IDENTICAL'), ('transactions_2026_09.jsonl', 'SKIPPED_IDENTICAL')]
Flagged for review: 21/92 descriptors, 15,033 transaction occurrences
Bills (candidates): 6,681; switches (candidates): 209
Bill evidence verified: PASSED | {'RECENT_CONSECUTIVE_MONTHS': 6561, 'HISTORICAL_CONSECUTIVE_ONLY': 0, 'MONTHLY_CANDIDATE_INSUFFICIENT_HISTORY': 120} | monthly-total outliers: 7
Switch evidence verified: PASSED | {'PROVISIONAL': 87,

'REPEATED_NEW_PROVIDER': 122,
'SUPPORTED_PAYMENT_PATTERN': 0} | externally confirmed: 0
Runtime 3.073s; peak RSS 70.7 MiB
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs | report: report/one_page_report.html
Rerun verification: outputs/workflow_rerun_verification.json
PASS: 4 CSVs byte-identical; 0 new batches

SUCCESS. Required CSVs are in outputs/; commercial + offline dashboard in report/.
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %
bash scripts/workflows/14_evaluate_pr_curves.sh --with-minilm

bash scripts/workflows/15_build_reports.sh --with-minilm

===== 14 | Exploratory PR curves, per-category AP and top-1 F1 =====
Loading weights: 100%|

| 103/103 [00:00<00:00, 3577.90it/s]
char_tfidf: top1_accuracy=0.789, macro_F1=0.635, macro_AP=0.698
minilm: top1_accuracy=0.754, macro_F1=0.657, macro_AP=0.654
Outputs: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/outputs/pr_evaluation
Results: outputs/pr_evaluation/

===== 15 | Offline commercial and inference-risk reports (read-only) =====
Loading weights: 100%|

| 103/103 [00:00<00:00, 7345.67it/s]
DASHBOARD: /Users/benjaminps/BWalker/Projects/UK JS/aptap-v1/report/risk_dashboard.html | 3 months | 92descriptors | MiniLM diagnostics=True | zero external resources
One-page commercial report: report/one_page_report.html
Interactive dashboard: report/risk_dashboard.html
(.venv) benjaminps@BenjaminPSs-MacBook-Pro aptap-v1 %

ChatGPT said:

You said:




Screenshot 2026...
Image

Screenshot 2026...
Image

those are the reports, lets explore the meaning on the classification and commercial risk for instance ADMIRAL insurance the risk is high but the policy is demo accepted. whereas aviva insurance the risk is also high and they both have an agreement, what is the diffference, what is the current policy. provide the traceability, which funtion, module, file generates it

Worked for 1m 54s

ChatGPT said:

You said:

are we ready to generate part d ?
Part D. Notes (about 1 hour, two pages maximum)

Write `NOTES.md` covering:

1. ****What you found wrong with the data****, and what you did about each problem.
2. ****What you don't trust**** in your own outputs, and why.
3. ****Running this in production.**** The pipeline and categoriser must run every month in two settings:
   - ****On ApTap's AWS****, for banks that send us their data.
   - ****Inside a bank****: 15 million customers, monthly files, no data can leave the bank's environment, you cannot install arbitrary software, and every release goes through the bank's change control. As volumes grow, file uploads into the bank's environment start timing out.

   Sketch both. What would you change in your design, and what would you do in your first week? Include:
   - ****Scale.**** Use the runtime and memory you measured to estimate what a 15-million-customer month needs, and say what

breaks first.
  - **Storage.** How you would model, partition and index the data at that size.
  - **Personal data.** What personal data is in these files, and how you would handle it: access, retention, and what may and may not leave each environment.
  - **Monitoring and cost.** How you would know if the categoriser started getting worse, and what it would cost per million transactions.
4. **How you used AI tools.** A short summary. The full chats go in your submission (see below).

Your AI chats

We encourage AI tools, and we want to see how you used them. **Include every LLM conversation you had while doing this task.** A submission without them is incomplete.

- **Chat apps** (Claude, ChatGPT, Gemini and similar): send a share link that opens the full conversation. Please also include an exported copy (PDF, Markdown or text), because links can expire.
- **Coding agents and IDE assistants** (Claude Code, Codex, Cursor, Copilot Chat and similar): export the session history or transcript files for this task and include them in the repository or zip. Use whatever export your tool provides.
- **If you did not use AI**, say so in `NOTES.md`.

What to send back

A Git repository link or a zip containing:

- your code, with a short section in the README on how to run it end to end, including with an extra file. A Git repository with your real commit history is preferred over a zip.
- `clean_transactions.csv`, `rejected_rows.csv`, `recurring_bills.csv`, `switches.csv`
- your report (Part C)
- `NOTES.md`
- your AI chats

How we assess it

In order of importance:

1. Whether your cleaning and features are right.
2. Whether your pipeline runs first time, re-runs safely and could go into production.
3. Whether your analysis and report are sound and honest about uncertainty.
4. Whether your notes show good judgement about data you can't fully trust.

A shorter submission with honest notes beats a longer one that hides its gaps.

Questions about the task: reply to the email it came with. Asking a clarifying question counts in your favour. We give the same answer to every candidate who asks the same question.

Worked for 1m 24s

ChatGPT said:

You said:

part d also demans a design sketch on a cloud architecture implementation in 2 scenarios, if the whle engine is in AATap AWS env or the other if the whole application is inside the customer network. that is a cloud engineering approach (Page 2) can you generate a depurated chat history in a pdf file witj a prper guided prompt that aligns with our current design. I want to provide the chat history as a pdf since my original chat hostory has some deviations so I want a coherent chat gpt conversation but in a pdf format output

Worked for 3m 31s

You said:

can I create a synthetically chat history in gpt based on a md file that contains all the knowledge base so I CAN SHOW THE THOUGHT PROCESS IN A sorted and intuitive manner

Worked for 6m 31s

ChatGPT said: